PHARMACOLOGY

Emetic and Antiemetic Drugs

LG 2
Dr. Begard O. Berzinji
GIT BLOCK

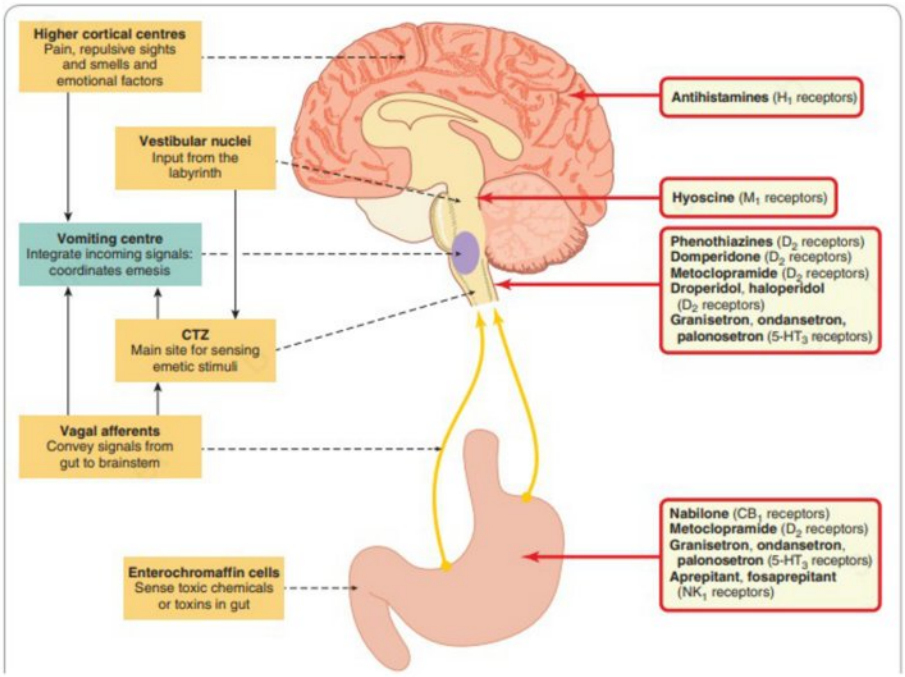
Pharmacology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which four neurotransmitter receptor types are involved in triggering vomiting?
2. Why is domperidone often preferred over metoclopramide in a patient with Parkinson's disease?
3. Which three-drug combination does the ESMO guideline recommend for high-emetic-risk chemotherapy?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Vomiting pathway & neurotransmitters



Vomiting-control pathway — receptor sites targeted by each antiemetic drug class (adapted from lecture slide 4).

Four neurotransmitter/receptor systems drive the vomiting reflex: **serotonin (5-HT3)**, **acetylcholine (muscarinic, M1)**, **dopamine (D2)**, and **histamine (H1)**. The chemoreceptor trigger zone (CTZ) senses emetic stimuli (drugs, toxins, metabolic signals) and relays to the vomiting centre, which also receives vestibular, GI vagal, and cortical input.

2 Antiemetic drug classes — site of action

Class	Drugs	Site of action	Comments
Antihistamines	Cinnarizine, cyclizine, promethazine	H1 (CNS, sedation) ± anticholinergic action in vestibular apparatus	Effective regardless of cause of emesis
Antimuscarinics	Hyoscine	Anticholinergic action in vestibular apparatus ± elsewhere	Mainly motion sickness
Cannabinoids	Nabilone	Probably CB1 receptors in GI tract	CINV where other drugs ineffective
Dopamine antagonists	Phenothiazines (prochlorperazine, perphenazine, trifluoperazine, chlorpromazine)	D2 receptors in CTZ	CINV, PONV, radiation sickness
	Droperidol, haloperidol	D2 receptors in GI tract	CINV, PONV, radiation sickness
	Metoclopramide	D2 receptors in CTZ and GI tract	PONV, CINV

Glucocorticoids	Dexamethasone	Probably multiple sites, incl. GI tract	PONV, CINV; typically combined with other drugs
5-HT3 antagonists	Granisetron, ondansetron, palonosetron, dolasetron, tropisetron	5-HT3 receptors in CTZ and GI tract	PONV, CINV
NK1 antagonists	Aprepitant, fosaprepitant, rolapitant, netupitant	NK1 receptors in CTZ, vomiting centre ± GI tract	CINV; given in combination

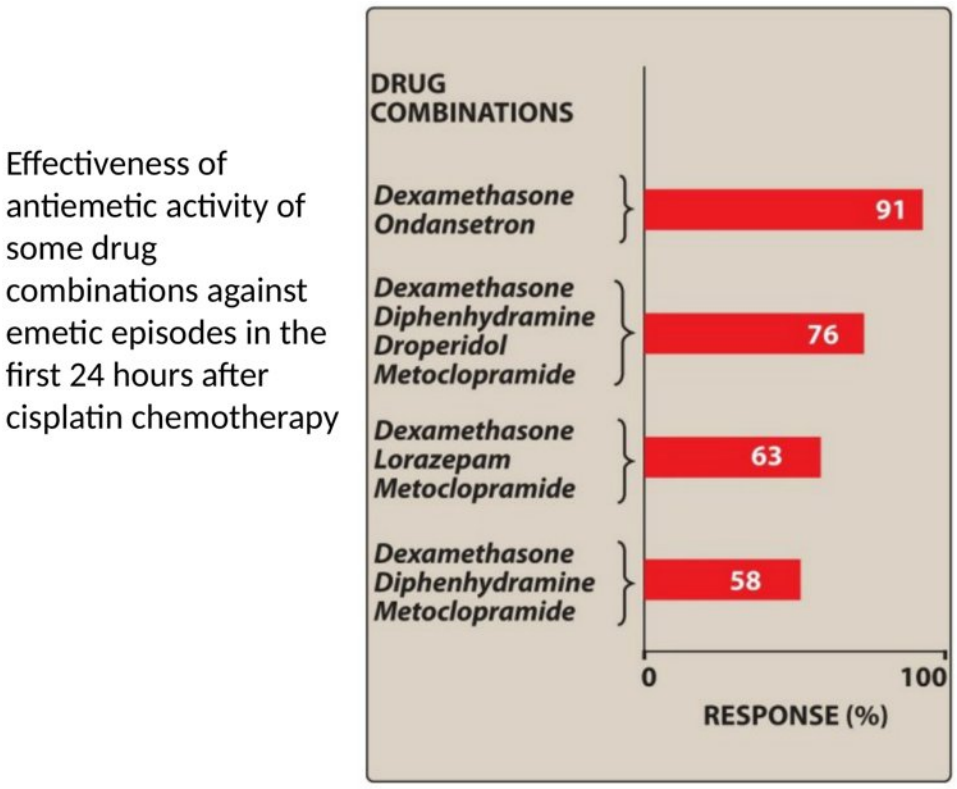
CINV = cytotoxic-drug-induced vomiting; PONV = postoperative nausea & vomiting.

3 Key agent-specific facts

5-HT3 antagonists	Act at CTZ (±GI tract). t½ 4–9h, palonosetron ~40h. No renal dose adjustment, but reduce dose in liver impairment. ADRs: headache, constipation, QT prolongation (most with dolasetron).
Metoclopramide	D2 antagonist + peripheral prokinetic. 10–20mg q6h; high first-pass metabolism; excreted in urine & breast milk. ADRs: EPS (movement disorders, torticollis, oculogyric crises — more in children/young adults), prolactin release.
Domperidone	Poor BBB penetration → fewer central effects; useful in Parkinson's. Small cardiac risk (dose/age-related, use restricted). t½ 4–5h, urinary excretion.
Phenothiazines & butyrophenones	D2 antagonists at CTZ; also block H1/M1. ADRs: sedation (esp. chlorpromazine), hypotension, EPS/ tardive dyskinesia.
NK1 antagonists	Block substance P. Aprepitant: CYP3A4 substrate & inducer (also CYP2C9 inducer) — avoid strong CYP3A4 inhibitors/inducers. Rolapitant: moderate CYP2D6 inhibitor. Fosaprepitant = IV prodrug of aprepitant. ADRs: fatigue, diarrhoea, hiccups.
Antihistamines	Cinnarizine, cyclizine, promethazine — good for motion sickness/irritants, not for direct CTZ stimulants. Promethazine: severe morning sickness.
Hyoscine	Oral or transdermal; dry mouth, blurred vision; less sedating (poor CNS penetration).
Nabilone	t½ ~120min. ADRs: drowsiness, dizziness, dry mouth, mood changes, postural hypotension. Effect reversed by naloxone.

4 Chemotherapy-induced vomiting — risk-based regimens

Emetic risk	ESMO-recommended regimen
High	Triple therapy: 5-HT3 antagonist + NK1 antagonist + dexamethasone (± olanzapine)
Moderate	Dual therapy: 5-HT3 antagonist + dexamethasone (± NK1 antagonist/olanzapine)
Low	Single agent: dexamethasone or a 5-HT3 antagonist
Minimal	No routine prophylaxis



Response rates of combination regimens against cisplatin-induced emesis — dexamethasone+ondansetron (91%) outperforms triple/dual combinations without a 5-HT3 antagonist.

5 Clinical uses by drug

Drug	Main clinical use(s)
Cyclizine	Motion sickness, vestibular disorders, postoperative/narcotic-related N&V
Cinnarizine	Motion sickness, vestibular disorders (e.g. Ménière's disease)
Promethazine	Severe morning sickness of pregnancy, motion sickness, vestibular disorders
Hyoscine	Motion sickness
Prochlorperazine	Migraine, vestibular disorders, radiation sickness, viral gastroenteritis, severe morning sickness
Metoclopramide	Migraine, radiation sickness, GI disorders, cytotoxics, postoperative prevention
Domperidone	Fewer CNS effects in Parkinson's patients (poor BBB penetration)
Ondansetron	Cytotoxics, radiation sickness, postoperative N&V
Nabilone	Cytotoxics (when other drugs ineffective)
Fosaprepitant	Cytotoxics (IV, combined with another drug)

P Must-remember pearls

- Apomorphine has a CNS depressant effect — contraindicated in respiratory failure.
- Ipecacuanha is contraindicated in kerosene/corrosive poisoning, unconscious patients, and morphine poisoning.
- Domperidone barely crosses the BBB → fewer central (extrapyramidal) effects, but carries a small cardiac risk, especially at high dose or in the elderly.
- Metoclopramide's central D2 blockade causes movement disorders (more common in children/young adults) and raises prolactin (galactorrhoea, gynecomastia, impotence, menstrual disorders).
- Among 5-HT3 antagonists, dolasetron carries the most notable QT-prolongation risk; palonosetron has by far the longest half-life (~40h vs 4–9h).
- Aprepitant is a CYP3A4 substrate (and an inducer of CYP3A4/CYP2C9); rolapitant is a moderate CYP2D6 inhibitor instead — avoid strong CYP3A4 inhibitors/inducers with aprepitant.
- Nabilone's antiemetic effect is antagonised by naloxone, implying opioid receptors contribute to its mechanism.
- ESMO high-risk chemotherapy → triple therapy (5-HT3 antagonist + NK1 antagonist + dexamethasone) ± olanzapine; moderate risk → dual therapy; low risk → single agent.

Q Test yourself

1. A patient receiving cisplatin develops delayed-phase chemotherapy-induced vomiting. Which receptor class, targeted by aprepitant, is particularly effective for this late phase?

A. NK1 (neurokinin-1) receptors
B. 5-HT3 receptors
C. D2 receptors
D. H1 receptors
E. M1 (muscarinic) receptors

2. A Parkinson's disease patient needs an antiemetic for GI symptoms. Why is domperidone preferred over metoclopramide in this patient?

A. Domperidone is a 5-HT3 antagonist, which does not affect dopamine pathways
B. Domperidone does not readily cross the blood–brain barrier, so it is less likely to worsen Parkinsonian symptoms
C. Domperidone has a much longer half-life, requiring once-daily dosing
D. Domperidone is excreted unchanged by the kidney, unlike metoclopramide
E. Domperidone blocks NK1 receptors instead of D2 receptors

3. A woman has severe morning sickness of pregnancy requiring drug treatment. Which H1 antagonist does the lecture name specifically for this indication?

A. Cyclizine **B.** Cinnarizine **C.** Promethazine **D.** Hyoscine **E.** Nabilone

4. A patient on high-dose metoclopramide for chemotherapy-induced nausea develops acute dystonia and restlessness. What is the mechanism behind these unwanted effects?

A. Metoclopramide blocks NK1 receptors in the basal ganglia
B. Metoclopramide is a direct GABA antagonist, causing excitotoxicity
C. Metoclopramide blocks 5-HT3 receptors in the vestibular nuclei
D. Metoclopramide blocks dopamine D2 receptors elsewhere in the CNS, beyond the CTZ
E. Metoclopramide induces hepatic CYP3A4, causing drug accumulation

5. A patient's antiemetic regimen combines a 5-HT3 antagonist, dexamethasone and an NK1 antagonist. According to the ESMO guidelines slide, which emetic-risk category of chemotherapy is this triple therapy intended for?

A. Minimal-emetic-risk chemotherapy
B. Low-emetic-risk chemotherapy
C. Moderate-emetic-risk chemotherapy
D. Any risk category, as first-line therapy
E. High-emetic-risk chemotherapy

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — Aprepitant blocks NK1 receptors in the CTZ and vomiting centre, and is effective in controlling the late (delayed) phase of cytotoxic-drug emesis.

2 — B — Domperidone does not readily penetrate the blood-brain barrier, so central (extrapyramidal) side effects are less likely — important in Parkinson's disease, where central D2 blockade would worsen symptoms.

3 — C — Promethazine is used for severe morning sickness of pregnancy (and was used by NASA for space motion sickness).

4 — D — Metoclopramide also blocks dopamine receptors elsewhere in the CNS, producing movement disorders, restlessness, torticollis, and oculogyric crises.

5 — E — Triple therapy (5-HT3 antagonist + NK1 antagonist + dexamethasone) is the ESMO recommendation for high-emetic-risk chemotherapy; moderate risk uses dual therapy, low risk a single agent.

SPACED REVIEW

☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 21 ☐ Pre-exam